



Modeling of Shock Propagation in Non-uniform Density Media

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The propagation of a normal shock wave in non-uniform density fields is studied with both theoretical methods and shock-resolving/shock-capturing simulations. The classical Chisnell-Whitham (C-W) theory is used to predict shock propagation in various density fields with 1D variations. The numerical results show that for oscillatory density profiles, the numerical results deviate from the theoretical solutions and exhibit additional long-wavelength oscillations, which are shown to be related to the re-reflected waves. A C-W model with re-reflected waves (CWRW) is proposed to compensate for the deviation. The results obtained by the proposed model show significant improvement over the classical C-W prediction of shock wave propagation in fluctuating density fields. The three-dimensional (3D) effects concerning the shock propagation in 3D non-uniform density media are studied by considering the ratio of the streamwise and spanwise length scales of the density field. The limiting behaviors for the wavenumber ratio approaching zero and infinity are investigated and used to provide bounds for the shock propagation in 3D non-uniform media.

I. Introduction

When a normal shock wave propagates through non-uniform media, the strength and geometry of the shock wave, as well as the structure of media itself, are modified through their interactions. This is an important fundamental problem relevant to Inertial Confinement Fusion (ICF), supersonic and hypersonic combustion, astrophysics, and other applications [1–4]. In an early theoretical study, Chisnell [5] investigated the one-dimensional propagation of shock waves through a non-uniform (variable density) media and derived an expression for the shock Mach number as a function of time. Later, Whitham [6] developed a more general analytical solution for a media with both varying density and pressure. These earlier formulations of shock propagation, even though useful, suffer from a number of difficulties, one of which originates from the fact that the waves generated from the post-region, i.e. the re-reflected waves, are neglected. The re-reflected waves effects are accounted for in the study conducted by Bird [7], who performed numerical simulations with the method of characteristics and compared them with the Chisnell-Whitham (C-W) results and asymptotic solutions. Soukhomlinov et al. [8] further examined the effects of temperature gradients on shock propagation. Their results were in good agreement with Bird's results. The above approaches can be applied to arbitrary non-uniform flows, while in the limit of weak fluctuations, this problem can be linearized to obtain the Linear Interaction Approximation (LIA). LIA is based on Kovaznay's decomposition [9] of the pre-shock non-uniform/turbulent fields into independent modes (acoustic, vortical and entropy) that propagate at their characteristic speeds. The shock propagation and post-shock flow statistics can then be theoretically derived from the linearized Rankin-Hugoniot (RH) jump conditions. The first theoretical solutions in the linear limit were developed by Moore [10], Ribner [11], which focused on the interaction between shock wave and pre-shock velocity disturbances. To study the effects of pre-shock non-uniform density field, McKenzie and Westphal [12] considered the interaction of isobaric entropy waves with an oblique shock. Mahesh et al. [13] then treated the 3D density fluctuations as combinations of linear entropy waves and conducted a linear analysis on shock-entropy wave interactions. Griffond [14], Huete et al. [15] have considered the pre-shock non-uniform density field as a mixture of gases with different molecular weights and extended the linear

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analysis to multi-component flows. These LIA results are capable of predicting both unsteady shock propagation and post-shock fluctuations, even though most of the studies have been focused on the later.

The canonical Shock-Turbulence Interaction (STI) problem is an example of shock propagation in non-uniform media, and has been the focus of most numerical studies in this area. In this case, the non-uniformity in the pre-shock region is mainly due to turbulent motions. In this type of problem, the shock propagation can reach a quasi-steady-state condition for a range of parameters, so that the statistical behavior of turbulence through the shock or in the post-shock region can be studied. Such a flow has been studied using shock-capturing and shock-resolving simulations for a range of flow and shock parameters [e.g. 16–21]. In contrast, there have been only a few numerical studies on shock propagation in non-uniform density fields. Mahesh et al. [13], Jamme et al. [22] studied the interaction of a fluctuating pre-shock vorticity-entropy field with a normal shock wave, which is related to shock/turbulent boundary layer interaction. Recently, our group investigated the shock interaction with multi-fluid variable density turbulence, and has shown that shock structure, shock intensity, and post-shock turbulence are strongly affected (figure 1 a,b) by the non-uniformities in density, even at low Atwood number (A_t) [23–25]. By further increasing the density ratio, the interaction results in a broken shock regime. Such a regime is also seen at high turbulent Mach numbers (M_t) [19] (figure 1 c). In previous studies, the interaction was studied in regimes where the shock motion was negligible [20] or the constant back-pressure outflow condition was used to stabilize the shock motion even at high M_t values or density ratios [19] so that the statistical analysis could be carried out. In this study, the transient shock propagation in variable density flow without boundary effects is of interest, as it resembles the conditions considered in the early theoretical work and practical applications like ICF. We aim to test the available theories on predicting the shock wave motion and post-shock flow under various conditions using accurate flow-resolving simulations with both shock-resolving and shock capturing approaches. We then extend the existing theories to account for the effects of re-reflected waves for 1D problems, with the hope of laying the theoretical foundation for more complicated 3D problems.

The rest of the paper is organized as follows. The simulation details are presented in section II. In this study, we conduct both shock-capturing and shock-resolving simulations and assess their accuracy in predicting the shock motion. The theoretical approach Chisnell-Whitham theory is then described in section III. In section IV, the numerical results are compared with the theoretical results for various types of non-uniform density flows. The main findings and conclusions are summarized in the last section.

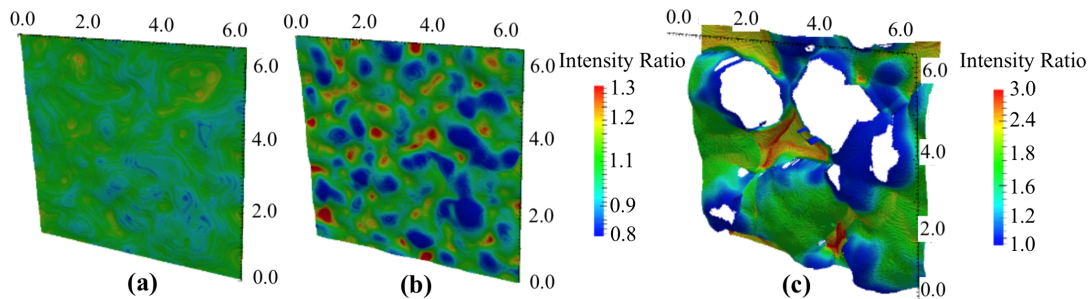


Fig. 1 Shock front in wrinkled shock regime (a,b) and broken shock regime (c). Shock intensity is represented by density jump $\Delta\rho$. Shock intensity ratio is calculated as $\Delta\rho/\Delta\rho_0$.

II. Numerical Methodology

The numerical investigation of the shock propagation in non-uniform density media are conducted by solving the dimensionless multi-component compressible Navier-Stokes equations, together with the perfect gas law for thermodynamic closure. The fluid viscosity is modeled using power law by $\mu = T^{0.75}$. The specific heat ratio $\gamma = C_p/C_v$ is assumed to be constant and equal to 1.4 for all components in the flow. The Prandtl and Schmidt numbers are $Pr = Sc = 0.75$.

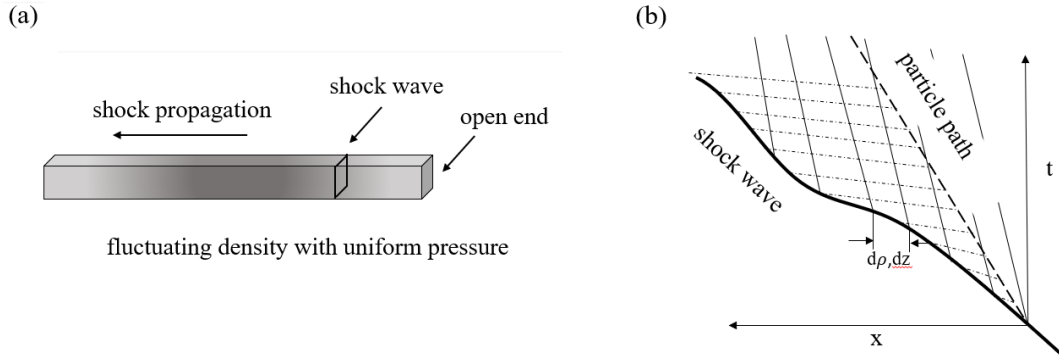


Fig. 2 (a) Numerical setup of the 1D configuration for shock-capturing and shock-resolving simulations. (b) Space-time diagram of the 1D shock propagation (thick solid line) in non-uniform density media. Due to the interaction between the shock wave and density non-uniformity, a reflected/forward-going wave is generated (thin solid line), which then interacts with the post-shock non-uniform density media at the intersection with the particle path (dashed line). As a result, a re-reflected/backward-going wave is generated (dotted line) and collides with the shock wave.

A. Shock-Capturing Simulation

For shock-capturing simulations, the hybrid Monotonicity-Preserving (MP)-Compact scheme developed by [24, 26] is used to solve the conservative form of the dimensionless compressible N-S equations. More specifically, the inviscid fluxes are computed by the fifth-order MP scheme as described in Li and Jaber [26] and the viscous fluxes are computed using sixth-order compact finite differences. The 3rd-order Runge-Kutta scheme is used for time advancement. The accuracy of the scheme has been verified in Li and Jaber [26] for various types of compressible flows and in [23, 24] for STI.

B. Shock-Resolving Simulations

Shock-resolving simulations are conducted by computing both the viscous and inviscid fluxes using sixth-order compact finite differences. To ensure numerical stability and minimize aliasing errors, the viscous terms are solved in the non-conservative form and the inviscid fluxes are solved in the skew-symmetric form [27]. A newly developed stable six-order compact scheme with a fifth order boundary scheme is implemented for the calculation of first-order derivatives [28]. The nonconservative form of the viscous fluxes contains the second-order derivative of primitive variables, which is calculated using the six-order compact scheme with a fourth order boundary scheme as described in Lele [29]. To fully resolve all flow scales, including shock width (δ), the mesh is further refined around the shock region. At least 12 grid points are used in the streamwise direction to resolve the shock profile, in contrast to the shock-capturing simulations, where the shock wave is numerical and is represented by only 2-3 grid points.

C. Simulation Setup

The numerical methods discussed above were developed and tested for 3D simulations. In this study, both 1D and 3D simulations are considered. The setup of the 3D simulations is similar to those considered in Tian et al. [23]. As shown in figure 2, the dimension of the computational domain are $(L_1, L_2, L_3) = (4\pi, 2\pi, 2\pi)$. As shown in figure 2 (a), the dimension of the computational domain is 4π in the streamwise direction. The normal shock wave is initialized in the middle of the domain at $x = 2\pi$. For the shock-resolving simulation, the shock is initialized using the laminar shock profile as derived in Thompson [30]. A moving reference of frame at the same speed as the laminar shock wave at reference conditions is used to avoid large movements of the shock and the need for excessively large computational domains. A buffer layer in the streamwise direction is set at the open end of the domain. To avoid the contamination of the shock motion by the boundary conditions, the simulations are stopped before the outflow acoustic waves propagate to the shock wave.

At the inflow, a non-uniform density field is superposed on a uniform mean flow that matches the laminar pre-shock condition for a shock wave with shock Mach number M_s . Here, we use the overbar $\bar{f}$ to denote the mean and f' to denote the fluctuations of the instantaneous turbulent field f . The density non-uniformity ρ' is introduced by correlating the mole fraction of the heavy fluid in the binary mixture to a scalar field. Different types of density non-uniformity are considered and discussed in the results section. To quantify the length scale and magnitude of the non-uniform density field, we use the peak wave number of the density spectrum, k_s , and Atwood number, $A_t = (MW_2 - MW_1)/(MW_2 + MW_1)$, where MW_1 and MW_2 are the molecular weights of the species in the binary mixture.

III. Chisnell-Whitham Theory

When the density non-uniformity is small, the LIA can be used to predict the shock motion. The interaction between non-uniform density media and a shock wave in the linear regime has been studied by considering the pre-shock fluctuations as either entropy [13] or compositional fluctuations [14, 31]. However, in many of the engineering applications, A_t or the magnitude of the density non-uniformity is too large, which invalidates the LIA assumptions. For example, the density fluctuations $\rho'/\bar{\rho} \sim 1.0$ in the hydrogen/air mixture in the scramjet engine. In the nonlinear regime, a different approach is needed to predict shock propagation in non-uniform density media. Early theoretical studies by Chisnell [5], Whitham [6] have considered the large density fluctuations as a sequence of weak contact discontinuities and derived a relation between the density change of $d\rho$ and the shock strength response dz , where z is the pressure ratio across the shock wave. The underlying assumption is that the density change across each contact discontinuity is weak so that the shock remains in quasi-equilibrium.

Figure 2 (b) shows a wave diagram of the shock propagation in a non-uniform density medium. As the shock wave encounters a density change $d\rho$, its Mach number and propagation speed are modified. First, a reflected/forward-going wave is generated (marked as thin solid line), which then interacts with the post-shock non-uniform density media at the intersection with the particle path (dashed line). As a result, a re-reflected/backward-going wave is generated (dotted line), which propagates in the same direction as the shock wave and eventually collides with the shock wave.

Chisnell [5], Whitham [6] derived the relation between upstream density jump $d\rho_u$ and dz as:

$$\frac{1}{\rho_u} \frac{d\rho_u}{dz} = \frac{2}{z-1} - \frac{1}{\lambda^2 + z} + \frac{2}{z-1} \left(\frac{1 + \lambda^2 z}{z(1 + \lambda^2)} \right)^{1/2} \quad (1)$$

where $\lambda = \sqrt{(\gamma - 1)/(\gamma + 1)}$ is a constant. In equation 1, the re-reflected waves are neglected. The differential equations can be integrated to obtain an explicit relation between shock strength and density.

$$\rho = A \frac{(z-1)^2}{(\lambda^2 + z)} \left\{ \frac{\sqrt{\lambda^2 + 1/z} + \lambda}{\sqrt{\lambda^2 + 1/z} - \lambda} \right\}^{2\lambda/\sqrt{1+\lambda^2}} \left\{ \frac{\sqrt{\lambda^2 + 1} - \sqrt{\lambda^2 + 1/z}}{\sqrt{\lambda^2 + 1} + \sqrt{\lambda^2 + 1/z}} \right\}^2, \quad (2)$$

where constant A can be calculated from the initial condition. This equation can then be numerically integrated over any given density profile to obtain the temporal evolution of the post-shock fields. Therefore, within this approximation, the shock strength and shock propagation speed can be obtained and the temporal variations of shock position can then be predicted for a given density profile. Due to the fact that the dependence of shock strength on density is implicit, z needs to be numerically obtained using an iterative method. Then the shock position (X_s) can be obtained using the following integration:

$$\begin{aligned} X_s &= \int dX_s = \int u_s dt = \int a(X_s) M(X_s) dt \\ &= \int \sqrt{\gamma P / \rho(X_s)} \sqrt{\left(\frac{z(\rho(X_s)) - 1}{2\gamma} (\gamma + 1) + 1 \right)} dt. \end{aligned} \quad (3)$$

By integrating the above equations numerically, the shock position X_s as a function of t can be obtained.

Chisnell [5] also formulated the equations for reflected and re-reflected waves. The interaction between weak contact discontinuity with a shock wave generates a forward-going elementary acoustic wave that propagates away from the shock wave. Its magnitude can be expressed as [5]:

$$dz_f = 1 + \frac{dz}{z} \quad (4)$$

where dz can be related to the density jump through equation 1. When this elementary wave interacts with the post-shock non-uniform field, backward-going re-reflected waves are generated, which propagate towards the shock wave and change the shock strength. Chisnell [5] derived the magnitude of the elementary re-reflected wave as:

$$dz_b = 1 - \frac{1}{4} \alpha(z, z_0) \frac{1}{\rho(z, z_0)} \frac{\partial \rho(z, z_0)}{\partial z_0} \frac{dz}{z} dz_0, \quad (5)$$

$$\alpha(z, z_0) = \left(\frac{\rho(z, z_0)}{\rho(z, z)} \right)^{1/4}. \quad (6)$$

where z_0 is the shock strength at the particle path. Here, $\rho(z, z_0)$ can be calculated using the following equation [5]:

$$\rho^{-1/4}(z, z_0) = \rho^{-1/4}(z_0, z_0) - \frac{1}{4\gamma} \int_{x=z_0}^z \rho^{-1/4}(x, x) \frac{dx}{x}, \quad (7)$$

where $\rho(z, z)$ is the density immediately after the shock wave for a given shock intensity z and can be calculated using the formula $\rho(z, z) = \rho \frac{(\lambda^2 + z)}{1 + \lambda^2 z}$. Based on these equations, the magnitude of the forward- and backward-going acoustic waves can be calculated by the numerical integration of the above differential equations for a given density profile. In this study, we aim to test the applicability of these theoretical results to various types of problems with the hope of finding a universal and simple model for the shock propagation in non-uniform density field.

IV. Results and Discussion

In this section, the numerical results obtained from both shock-capturing and shock-resolving simulations are examined and compared with existing theoretical results. An improved model is then proposed to compensate for the missing nonlinear effects in these models. Finally, a study on shock propagation in 3D non-uniform density fields is conducted to understand the 3D effects.

A. Shock Wave Propagation in Media with Fluctuating Density

The fundamental understanding of shock propagation in fluctuating density fields is critical to the study of shock-turbulence/shock-boundary layer interactions. This problem provides a simplified configuration for the more complex 3D case with turbulence. Simplified pseudo-1D simulations are first conducted. The shock is initialized in a fluid with $\rho = 1.0$ at $x \approx \pi$; then a fluid with density profile given by $\rho = 1.0 + (1 - A_t)/(1 + A_t) \sin(k_s x)$ is superposed on the mean flow at the inlet. The propagation of shock wave into non-uniform density with a sinusoidal shaped profile is called the Shu-Osher problem [32].

In figures 3 (a,b), the temporal position of the shock is shown for a Mach 2 shock and relatively small (but not in linear regime) density ratio of 2:1 ($A_t = 0.33$) for two wavenumbers of $k_s = 1$ and 4. As seen in the figure, the C-W theory predicts a shock motion that resembles the sine wave variation in the density profile, but with a constant mean velocity relative to the unperturbed shock, so that the shock wave moves upstream. The match between the numerical and theoretical solutions is fairly good during the first (sine) wave cycle. However, after the first cycle, the shock wave acquires additional oscillatory motions instead of moving upstream like the original sine wave, as predicted by the theory. Thus, the current shock-capturing simulations are predicting long-wavelength oscillatory motions superposed on the original sinusoidal motion. Shock-resolving simulations are conducted to verify this observation. As seen in figure 3 (b), shock-resolving results match very well with the shock-capturing results, proving that the deviation from C-W solution and the long-wavelength oscillatory motions are indeed physical. This is not surprising though, as the scale separation criterion mentioned above is well satisfied. The ratio of the wavenumber, k_l , of the first long-wavelength harmonic of the shock motions to the sine wave wavenumber, k_s , is denoted as $r_l = k_l/k_s \approx 1/4$. This ratio appears to be the same for $k_s = 1$ and 4. To explain these long-wavelength motions, we consider two velocities that are related to the behavior of shock: 1) the velocity at which the pre-shock medium approaches the shock wave, v_p , and 2) the velocity of the re-reflected waves approaching the shock wave relative to the shock speed, v_r . Effectively, v_p carries the density fluctuations responsible for moving the shock wave, while v_r can be found from the local sound speed behind the shock.

The re-reflected waves can eventually collide with the initial shock and affect its strength and speed [7]. The ratio of v_r/v_p is denoted as r_v , and we find that $r_v \approx r_l$. For a weaker shock, as shown in figure 5(c), r_v and r_l are small in magnitude: $r_v \approx r_l \approx 1/12$. Thus, the early time agreement with the C-W theory is improved and the long-wavelength variation in shock position is also clearer because the larger separation of length scales between the short and long wavelength fluctuations. It can be inferred that the re-reflected waves, which are responsible for the long-wavelength motions of the shock [7], are also responsible for the deviation from the theoretical solution. Departures from the C-W prediction have been observed for linearly varying density profiles. However, for a fluctuating density profile, it seems that the re-reflected waves play a more important role in changing the shock strength and propagation, making it more different than from the C-W model.

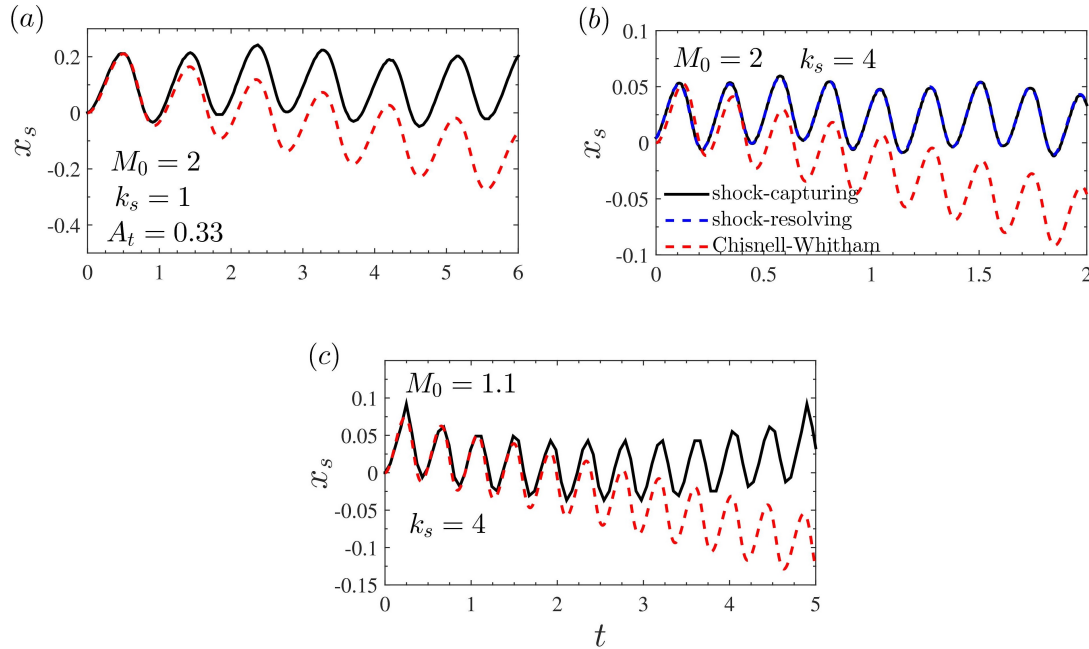


Fig. 3 Shock propagation in a fluctuating density field. (a,b) $M_0 = 2.0$, density ratio 2.0, and $k_s = 1$ and $k_s = 4$ and (c) $M_0 = 1.1$, density ratio 2.0, and $k_s = 4$.

On the other hand, the agreement of the short wavelength observed from the figure 3 indicates that C-W solution could provide good estimation for the short wavelength fluctuations. The magnitude of the short wavelength shock oscillation can be obtained by integrating equation 3 over the very first sin wave. The wavelength of the pre-shock density profile is fixed, so the results are only a function of shock Mach number M_s and Atwood number A_t . The shock position, obtained from both numerical simulations and numerical integration of C-W equation are compared in figure 4 for cases with different M_s . The C-W theoretical results show that as A_t increases, the shock oscillation becomes larger. Evidently, cases with lower M_s have a relatively larger shock oscillation. It can be seen that the numerical results agree with the theoretical results. Thus, C-W solutions can reasonably predict the short wave-length shock oscillations in the nonlinear regime.

B. C-W Solution with Correction for Re-reflected Waves

To further extend the applicability of the C-W theory to more complex problems, the effects from the re-reflected waves need to be considered. Chisnell [5] has considered the collision between the shock wave and the re-reflected waves for the linearly increasing density profiles. However, the suggested equations are not closed, making it difficult to calculate the shock strength and shock propagation, without resorting to numerical simulations as expensive as the original N-S/Euler solver. In this section, we propose an approximation of these equations that can be used as a theoretical model for the estimation of the re-reflected wave effects on the shock propagation in fluctuating density fields.

Following Chisnell [5], the magnitude of the generated acoustic waves immediately after the shock wave can be

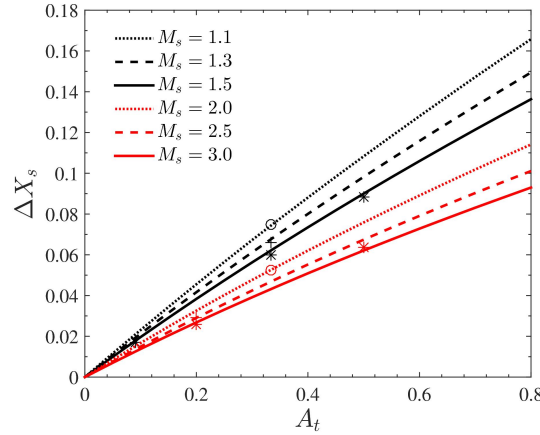


Fig. 4 Comparison of the shock position fluctuations between numerical results and theoretical results. Results from shock-resolving simulations are marked using symbols for $M_s = 1.1$ (black circle), $M_s = 1.3$ (black cross), $M_s = 1.5$ (black star), $M_s = 2.0$ (red circle), $M_s = 2.5$ (red cross), $M_s = 3.0$ (red star).

estimated by numerically integrating equation 2 as:

$$z_f(X_s) - 1 = \int_{z_{init}}^{z(X_s)} \frac{dz}{z} \quad (8)$$

Similarly, the magnitude of the re-reflected waves can be estimated by integrating equation 5 over dz_0 as:

$$z_b(X_s) - 1 = - \int_{z_{0,init}}^{z_0(X_s)} \frac{1}{4} \alpha(z, z_0) \frac{1}{\rho(z, z_0)} \frac{\partial \rho(z, z_0)}{\partial z_0} \frac{dz}{z} dz_0 \quad (9)$$

From these equations, the profiles of the numerically integrated waves can be estimated.

In order to decompose the post-shock waves in the numerical simulations, a wave decomposition method proposed by Payri et al. [33] is used. The forward propagating waves (p_f) or backward propagating waves (p_b) can be calculated using:

$$p_f = p_0 \left[\frac{1}{2} \left(1 + \left(\frac{p}{p_0} \right)^{\frac{\gamma-1}{2\gamma}} \left(1 + \frac{\gamma-1}{2} \frac{u_1}{a} \right) \right)^{\frac{2\gamma}{\gamma-1}} \right] \quad (10)$$

$$p_b = p_0 \left[\frac{1}{2} \left(1 + \left(\frac{p}{p_0} \right)^{\frac{\gamma-1}{2\gamma}} \left(1 - \frac{\gamma-1}{2} \frac{u_1}{a} \right) \right)^{\frac{2\gamma}{\gamma-1}} \right] \quad (11)$$

Therefore, based on the definition of wave strength z , z_f can be calculated by using $p_{f,max}/p_{f,min}$ and z_b by using $p_{b,max}/p_{b,min}$. The relative importance of the re-reflected waves can be estimated using the ratio of the magnitude of the two waves z_b/z_f .

The ratio z_b/z_f obtained from theory and numerical simulations is plotted in figure 5. The theoretical results predict that as the initial shock Mach number M_s increases, the ratio of z_b/z_f decreases. In addition, as the magnitude of density non-uniformity A_t increases, the re-reflected wave z_b also becomes less important. Numerical results, as marked in the figure by symbols, match very well with the theoretical results for both shock-capturing simulations and shock-resolving simulations. This proves that the post-shock re-reflected waves are correctly predicted in Ref. Chisnell [5].

Chisnell [5] proposed a correction to the shock propagation theory to compensate for the effects of the re-reflected waves. Thus, the strength of the re-reflected waves z_b can be incorporated into the original equation. Unfortunately, the equations are not closed. Here, we propose several simplifications to make the final equations tractable.

As shown in figure 2 (b), the interaction of the re-reflected wave generated by a forward-going wave dz_1 with the non-uniform density field generated by dz_0 can be expressed by equation 9. This wave propagates upstream and

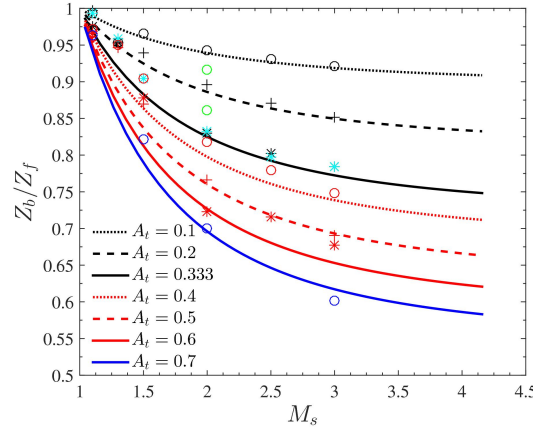


Fig. 5 Comparison of the relative magnitude z_b/z_f of the re-reflected waves. Results from shock-resolving simulations are marked using symbols for $A_t = 0.1$ (black circle), $A_t = 0.2$ (black cross), $A_t = 0.333$ (black star), $A_t = 0.4$ (red circle), $A_t = 0.5$ (red cross), $A_t = 0.6$ (red star), $A_t = 0.9$ (blue circle).

eventually changes the shock intensity at dz . Therefore, the modification of shock strength dz is not only affected by the pre-shock density change $d\rho$, but also is affected by the re-reflected waves along the negative characteristic line, which is marked as the dash-dotted line in figure 2 (b). Therefore, we use dz_{re} to represent the modified shock wave strength after the interaction with the re-reflected wave. Using the magnitude of the re-reflected wave dz_{rere} that collides with the shock wave, dz_{re} can be estimated from:

$$\frac{d\rho_u}{\rho_u} = \left(\frac{2}{z-1} - \frac{1}{\lambda^2+z} + \frac{2}{z-1} \left(\frac{1+\lambda^2 z}{z(1+\lambda^2)} \right)^{1/2} \right) dz_{re} + \frac{4}{z-1} \left(\frac{1+\lambda^2 z}{z(1+\lambda^2)} \right)^{1/2} dz_{rere} \quad (12)$$

The only remaining unknown is the value of dz_{rere} along the negative characteristic line. In Chisnell's theory [5], the magnitude of the re-reflected wave which catches up the incident shock is obtained from the following equation:

$$1 - dz_{rere} = 1 - dz \int_{z_0=0}^{z_0=z} \frac{1}{4} \alpha(z, z_1) \frac{1}{\rho(z_1, z_0)} \frac{\partial \rho(z_1, z_0)}{\partial z_0} \frac{dz_1}{dz} \frac{1}{z_1} dz_0 \quad (13)$$

The unknown quantity $\frac{dz_1}{dz}$ represents the portion of the elementary re-reflected wave that catches up with and then modifies the initial shock wave. This can be determined by the relative speed of the shock wave and the re-reflected wave. The main assumption we use to calculate this quantity is that the post-shock acoustic speed is constant so that the characteristic lines are straight. This implies $\frac{dz_1}{dz} = \frac{u_b - u_s}{u_s} = \text{constant}$, where u_b and u_s are the velocities of re-reflected wave and shock wave in the laboratory frame. The above assumption is reasonable when the shock wave is propagating in a fluctuating density field with a fixed mean value. By making this assumption, one can integrate the re-reflected wave along the characteristic line without knowing the full post-shock field.

After obtaining the magnitude of the re-reflected wave, equation 12 can be solved for dz_{re} , and the modified shock strength $z_{re}(t)$. The modified shock position can then be calculated by replacing $z(t)$ with $z_{re}(t)$ in equation 3. The modified C-W solution will be referred to as C-W with re-reflected waves (CWRW).

Figure 6 compares C-W and CWRW predictions with the shock-resolving simulation. It can be seen that by including the effects of re-reflected waves, the CWRW model can significantly improve the prediction of shock propagation in sinusoidally shaped density field. Evidently, both the mean deviation from the C-W solution and the long-wavelength oscillation are predicted by the CWRW model.

C. Shock Propagation in 1D Random Density Media

The above analysis based on simplified density profiles can provide some basic assessment of the C-W and CWRW models. However, due to the nonlinear nature of the problem, the CWRW model solutions may not be accurate in the prediction of shock propagation in a realistic random density media. In this section, we will compare C-W and CWRW results for a randomly generated smooth density profiles that satisfy a given spectrum with the numerical results.

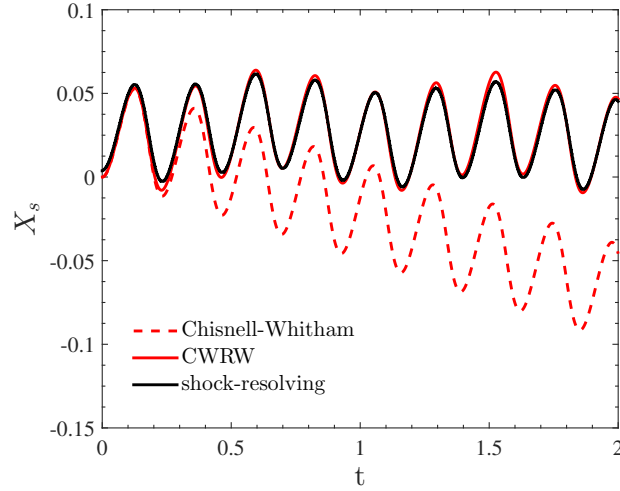


Fig. 6 Shock propagation in sinusoidally varying density field predicted by shock-resolving simulation, C-W theory and the proposed CWRW theory.

Here, we will use the same spectrum as used in Tian et al. [24] for the 1D density field. The characteristic length scale of the density field is modified by changing the peak wave number of the spectrum. The generated density field will then be used as the inflow condition for both numerical simulation and numerically integrated C-W and CWRW models. Figure 7 shows the shock propagation in two realizations of the random density field with $k_s = 4.0$ and $A_t = 0.5$. The numerical solutions are plotted using solid black lines and C-W solutions are represented as dashed lines. As observed, the C-W solutions are in overall agreement with the numerical simulations. However, they still gradually deviate from the numerical solution, the same as we have observed in figure 3 (b). We also note that the long wavelength oscillations are not visible. The CWRW model shows improvement over the C-W solution and agrees better with the shock-resolving simulation predictions. These results confirm that the CWRW model is superior to C-W model and is able to provide better predictions of the shock propagation in 1D random density field.

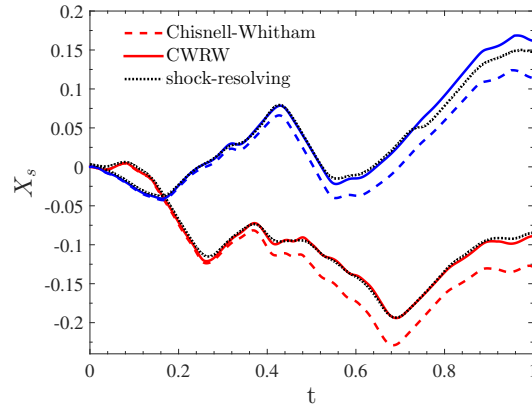


Fig. 7 Shock propagation in random density media for different realizations of the random density field, represented using different colors.

D. Shock Wave Propagation in Three-dimensional Non-uniform Density Media

In most applications, the non-uniform density distribution/media is three-dimensional. The interaction between small amplitude 3D density fluctuations with a normal shock wave has been theoretically studied by Griffond [14], Huete

et al. [15] using linear theory. These studies have been focused on the prediction of the post-shock field rather than the shock propagation and wrinkling. However, the amplitude of density fluctuations or A_t is usually too large in real-life applications, which violates the assumptions of these analyses.

In the above discussions, we have considered the ability of C-W and CWRW models to predict shock propagation in 1D non-uniform density media. The advantage of CWRW model is that it can be extended to nonlinear regime. When the 3D spatial effects in the spanwise directions play a role, one may not simply treat each point on the 3D shock as individual 1D field normal to shock. In this section, the 3D effects of shock propagation in non-uniform media are discussed.

1. 3D Sinusoidal Shaped Density Field

The shock propagation in a realistic 3D random non-uniform media is understandably complicated. To simplify the problem, here we consider a 3D sinusoidal shaped density field that satisfies $\rho(x, y, z) = 1.0 + A_t \sin(k_x x + k_y y + k_z z)$. By using this 3D density field, one can prescribe A_t and k_x, k_y, k_z to modify the intensity and length scale of the non-uniform density field. The anisotropy of the non-uniform density can also be changed by modifying the ratio of the streamwise to spanwise length scales $r_k = \frac{k_x}{k_{yz}}$, where $k_{yz} = \sqrt{k_y^2 + k_z^2}$. Using a coordinate transformation, one may show that the 3D sinusoidal shaped density field is essentially a 2D plan wave with A_t , k_x , and k_{yz} being the important parameters. When the wavenumber ratio r_k is equal to one, the density field becomes isotropic. When r_k approaches infinity, the density field becomes quasi-1D. On the other hand, when r_k approaches zero, the spanwise shock surface wrinkling dominates the shock propagation. Fundamental understanding of these two limits and how the shock behaves when the ratio r_k is varied are critically important to the modeling of shock propagation in 3D non-uniform media. The following results and discussions on the 3D effects are divided into two parts. The limit when scale separation ratio r_k approaches infinity is studied first, followed by an examination of the limit for r_k goes to zero.

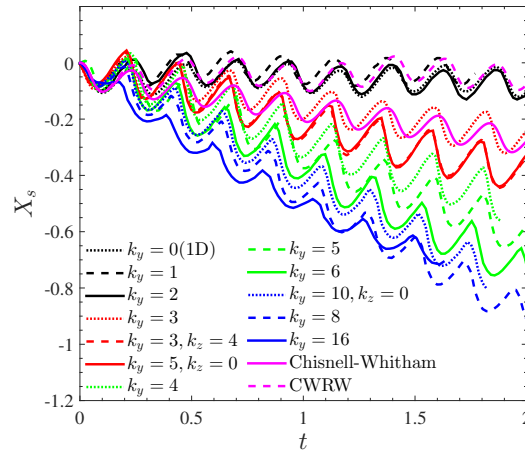


Fig. 8 Shock propagation in 3D sinusoidal shaped density media. The temporal development of shock position on certain point on the shock surface is plotted. $k_x = 4$ for all the cases and $k_y = k_z$ unless the values are indicated by the figure legend.

In figure 8, the instantaneous shock positions for several selected points on the shock surface are plotted as a function of time. We have verified that the selected points on the shock surface can represent the overall behavior of the shock wave, so the following discussions and conclusions may apply to the whole 3D shock surface. For all the cases considered in figure 8, A_t is very close to 0.5 and the length scale in the streamwise direction is fixed with $k_x = 4$. The 3D effects are studied by varying k_y and k_z . Unless otherwise stated, k_y is equal to k_z . The results from C-W and CWRW models are also plotted. Note that both models are for 1D shock propagation, so in this 3D case, we have made the assumption that the selected point on the shock wave is only affected by the non-uniform density field right before that point. This essentially implies that the corresponding point on the shock is moving like a 1D normal shock. When k_y, k_z are reduced to zero, the 3D simulation becomes quasi-1D. As expected, in this case, the predicted values fall onto the CWRW model curve, which deviates from C-W theory on average. As k_y, k_z are increased from 0 to 1 and 2, the

results do not show significant deviations from the quasi-1D ($k_y = k_z = 0$) case and CWRW model. This suggests that with a wavelength ratio greater than or equal to $\sqrt{2}$, the shock motion can still be approximated by the 1D theory and the 3D effects are generally negligible. Further increasing k_y, k_z to 3, however, significantly changes the behavior of the shock wave. In addition to the sinusoidal-shaped fluctuations, the shock wave moves forward with a mean velocity similar to that predicted by the C-W theory. However, this may be a coincidence, as for other wavenumbers the mean velocity is different than the C-W prediction. The sudden change in the shock behavior when the wavenumber ratio is decreased from $\sqrt{2}$ to $4/(3\sqrt{2})$ is hypothesized to be caused by the nonlinear effects induced by the 3D non-uniform density field. When the density field becomes more isotropic, the effects from the post-shock re-reflected waves seem to be weakened. Additionally, the 3D effects further increase the mean shock propagation speed when k_y, k_z are further increased. We have conducted similar simulations for $k_x = 1$ and 2 with same wavenumber ratio. The results (not shown) are found to be in good agreement with the results above, implying that the wavenumber ratio is the only important parameter.

To study the other (r_k goes to zero) limit, the length scales in the spanwise directions are further decreased by increasing k_y, k_z . By doing so, we observe a smooth increase in the mean shock speed until $k_y = k_z = 8$ or a wavenumber ratio of 0.35. Results from a smaller wavenumber ratio have almost the same mean propagation speed. This suggests that when the wavenumber ratio is less than 0.35, the mean shock propagation speed reaches another limit and in this limit, the density fluctuations in the streamwise direction no longer play a role on the mean shock propagation.

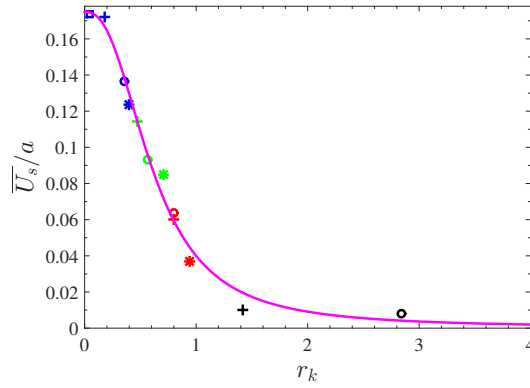


Fig. 9 The dependence of mean shock speed on wavenumber ratio for shock propagation in 3D non-uniform density media with $k_x = 4$.

To better show the dependence of the shock propagation speed to the wavelength ratio, all the results for $k_x = 4$ with varying k_y and k_z are plotted on figure 9. The shock speed in this figure is normalized using the mean sound speed a in the variable density media. A hyperbolic tangent function is fit to the data points using least square method. The two limits of mean shock propagation for wavelength ratios $r_k = 0$ and ∞ are not well captured by the fitted function but can be inferred from figure 8 (b). For the limit of r_k approaching infinity, the mean shock propagation speed reaches a small non-zero value, much smaller than that predicted by the C-W theory. This limit is essentially a pseudo-1D case, which can be numerically approached by setting $k_y = k_z = 0$. For this limit, the improved CWRW model, proposed earlier, gives a better prediction of the mean shock propagation speed than the C-W theory. For the other limit, results show that when r_k approaches 0, the shock propagation speed becomes independent of the wavelength ratio and reaches a maximum speed around Mach 0.175. In general, dimensional analysis show that the mean shock speed should be a function of A_t and M_s , i.e. $\overline{U}_s(A_t, M_s)$. For this limit, the smallest wavenumber ratio achieved in the current numerical simulations is 0.044. Unlike the other limit, one cannot simply set the streamwise wavenumber $k_x = 0$, because this will result in a highly distorted and broken shock wave. It is also problematic to continuously increase the spanwise wavenumber k_y, k_z in simulations, because by significantly decreasing the fluctuating density wavelength, the diffusion effects become dominant. In that case, the shock wave propagation speed will depend on Reynolds number Re . Previous simulations satisfy the scale separation criterion between the characteristic shock length scale and non-uniform density length scale, so the viscous effects can be neglected in the shock interaction. This essentially means that the above observations correspond to a Reynolds number being infinitely large across the shock. In this study, the limit $r_k = 0$ can only be approximated through simulations with $r_k = 0.044$ when there is a reasonably large length scale separation.

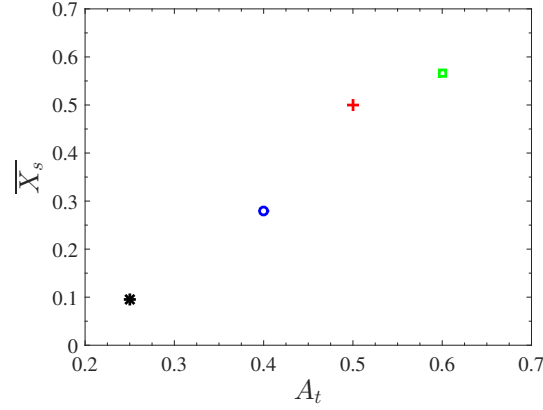


Fig. 10 The dependence of mean shock speed on Atwood number for shock propagation in 3D non-uniform density media with $k_x = 4$.

The dependence of $\overline{U}_s(A_t, M_s)$ on Atwood number at the limit of $r_k \approx 0$ is also studied with fixed M_s . The results are plotted in figure 10. We note that the mean propagation speed increases when the Atwood number is increased, which is mainly due to nonlinear effects. When $A_t = 0$, the shock is stationary and the mean shock propagation speed is zero. As the Atwood number increases, the nonlinear effects introduced through the non-uniform density field, as well as the spanwise shock wrinkling, will increase, causing the increase in mean shock propagation speed. In 1D cases (r_k approaches infinity), the re-reflected waves cancel out part of the nonlinear effects explained when we described the proposed CWRW model, making the shock propagation speed smaller than the 3D cases. The prediction of $\overline{U}_s(A_t, M_s)$ at the limit of $r_k \approx 0$ is still not clear and requires further investigation.

By removing the mean shock propagation from the instantaneous shock position, one can obtain the magnitude of fluctuations in shock position at different spanwise locations, which can be used to describe the shock surface wrinkling. Figure 11 shows the magnitude of the fluctuations in shock position as a function of the wavelength ratio ($L^2(\Delta X_s)$). A least-square-fit to the function $(c(1)r_k + c(2)) \times e^{c(3)r_k^2} - c(2)$ is also shown. Here we note that $L^2(\Delta X_s)$ peaks around $r_k \approx 0.57$ and continuously decreases as the wavelength ratio increases. The asymptotic limit at $r_k = \infty$, representing the 1D case, is not accurately captured by the fitted function but can be reasonably well predicted using the CWRW model. On the other hand, when the wavelength ratio decreases and approaches zero, the spanwise length scale of the non-uniform density field becomes smaller. Numerical results show that this will cause the shock oscillations to decrease. As the wavelength ratio approaches this limit, the fluctuations in the spanwise directions become more important to the shock wrinkling. This indicates that the decrease in length scale of the spanwise density fluctuations causes the decrease in the shock surface wrinkling.

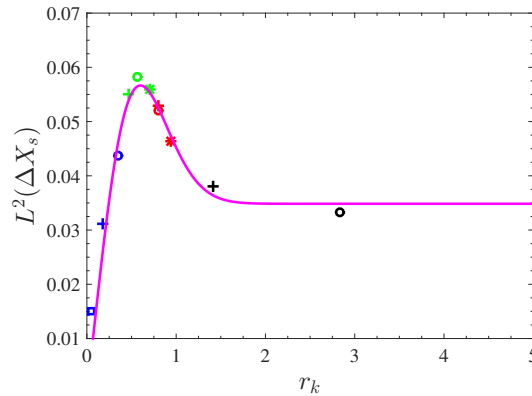


Fig. 11 The dependence of L^2 norm of the shock fluctuations on wavenumber ratio for shock propagation in 3D non-uniform density media with $k_x = 4$.

2. 3D Random Density Field

For a 3D random density field, the shock propagation becomes extremely difficult to predict, because there exists a wide range of spanwise and streamwise length scales. Despite the difficulty in the accurate prediction of shock position, the results above suggest that for a fixed A_t and M_s , the shock propagation speed should be bounded by the two limits described in the previous section. In this study, a shock-capturing simulation was also conducted to study shock propagation in a randomly generated density field. The generation of the 3D density field was described in the first chapter and is consistent with canonical shock-turbulence interaction studies [23]. Here, we keep the peak wavenumber of the density spectrum to be $k_s = 4.0$. The Atwood number is set to be 0.5.

Figure 12 shows the mean shock propagation for many individual points on the shock surface. The two bounds for the Atwood number of 0.5 and shock Mach number of 2 are also shown. Interestingly, the results show that for each individual point on the shock, the mean propagation is bounded by the two limits, even though the density field is generated randomly. This suggests that even though predicting the instantaneous shock surface response to the random density fluctuations is very challenging, one can still provide bounds for the shock position at every point on the shock surface.

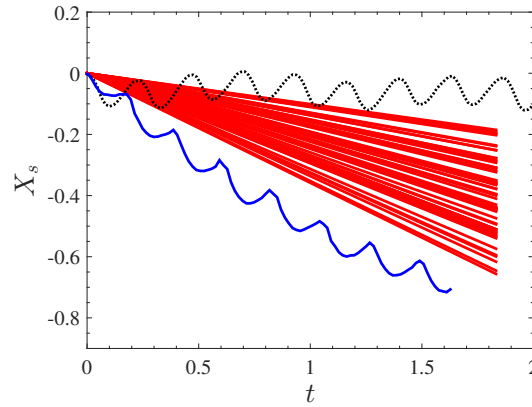


Fig. 12 The mean propagation of shock wave in 3D random density media. Red lines represent points on the shock surface. The blue and black curves represent the bounds corresponding to the two asymptotic limits.

V. Conclusions

In this study, accurate flow-resolving, shock-resolving/shock-capturing simulations are conducted together with theoretical analysis to study the shock propagation in non-uniform density media. In the nonlinear regime, the C-W theory [5, 6] gives explicit solutions for the shock propagation in 1D non-uniform density media by neglecting the post-shock field effects, i.e. the re-reflected waves. Pseudo-1D numerical simulations of shock propagation in oscillatory density profiles show that the numerical results deviate from the theoretical solutions and exhibit additional long-wavelength oscillations. This is then shown to be related to the re-reflected waves. The magnitude of the re-reflected waves is numerically obtained by extending the original Chisnell's formulation, which shows excellent agreement with the numerical results. A simplified model that includes the effects of re-reflected waves is then proposed to compensate for this error. The results obtained by the proposed model for the shock wave propagation in fluctuating density fields show significant improvement over those predicted by the classical C-W theory.

The shock propagation in 3D non-uniform density media is also studied, first using a simplified 3D sinusoidal shaped density profile. The 3D effects are investigated by varying the ratio of the streamwise and spanwise wavenumbers of the harmonic density field. The asymptotic limits for wavelength ratio approaching zero and infinity are used to bound the 3D effects on the shock propagation. It is shown that when the wavenumber ratio is larger than 1.4, the results are very close to those predicted by the proposed CWRW model, which implies one can treat each point on the shock wave as an independent 1D shock. Decreasing the wavenumber ratio results in an increasing shock propagation speed. When the wavenumber ratio approaches zero, the shock propagation speed reaches another asymptotic limit. Thus, the shock propagation speed is bounded by the two limits corresponding to r_k approaching 0 and infinity at any fixed Atwood number, A_t and shock Mach number, M_s . The shock wrinkling is also found to be dependent on r_k , reaching its

maximum strength at $r_k \approx 0.57$ for $A_t = 0.5$ and $M_s = 2$. Numerical study of shock propagation in 3D random density field indicates that the shock propagation speed on any point at shock surface is still bounded by the two limits observed in our 3D simulations with sinusoidal density profiles.

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